BDD Chawls Redevelopment Project, Worli, Mumbai

Details | Data Entry | Report

# Initial Vertical Load Test Report

Pile ID: TP-01 | Location: Building-3 Wing B
Test Date: 09 December 2025 | Test Method: IS 2911 (Part 4) - 2013

**Test Result: PASSED**
Net Settlement: 4.14mm (Limit: 12mm)

PASSED

**TEST LOAD**
**1050.0** MT
2.5× Design Load

**MAX SETTLEMENT**
**7.22** mm
At test load

**ELASTIC REBOUND**
**3.08** mm
Recovery after unloading

**NET SETTLEMENT**
**4.14** mm
Limit: 12mm (IS 2911)

**SAFE LOAD ADOPTED**
**420.0** MT
Per design criterion

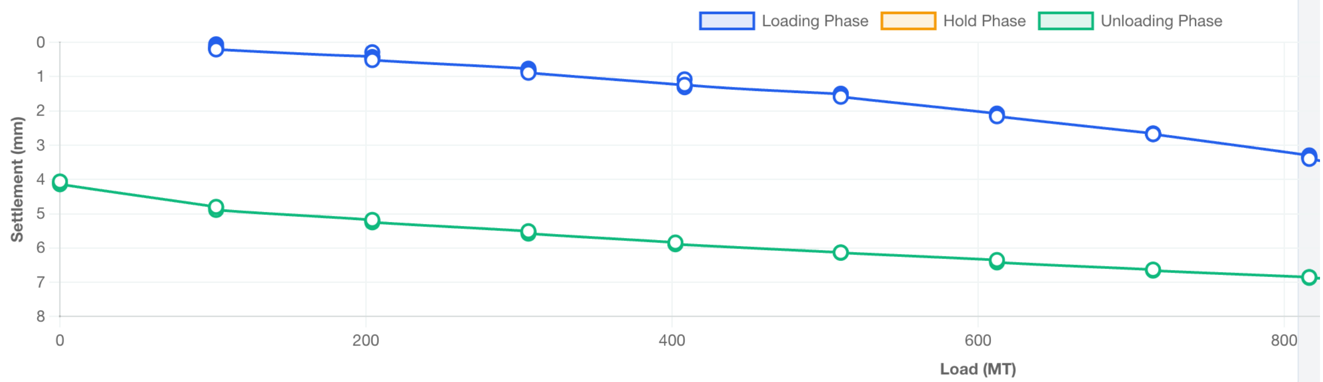
**DESIGN LOAD**
**420.0** MT
Working load capacity

## Pile Specifications

| | |
|---|---|
| Pile ID<br>**TP-01** | Diameter<br>**900 mm** |
| Depth<br>**11.51 m** | Concrete Grade<br>**M40** |
| Ram Area<br>**2551 cm²** | Jack Name<br>**-** |
| Dial Gauge LC<br>**0.01 mm** | Method<br>**IS 2911 (Part 4)** |

## Load vs Settlement Curve

Loading Hold Unloading



### 4.0 Results & Conclusion

AI Generated

Edit Regenerate

In accordance with the guidelines outlined in IS 2911 (Part 4) - 2013, the Initial Vertical Pile Load Test (IVPLT) for Pile ID TP-01 has yielded significant results. The maximum settlement measured during the test was 7.22 mm, with an elastic rebound of 3.08 mm, resulting in a net settlement of 4.14 mm. These values confirm that the net settlement is well within the acceptance criteria stipulated by the standard, which allows for a maximum settlement limit of 12 mm.

Given that the pile has successfully met these criteria and demonstrated a settlement well below the required threshold, it is concluded that the pile has PASSED the test. Consequently, the safe load that can be adopted for working piles has been set at 420 MT, which is consistent with the design load. Therefore, it is established that this pile is suitable for the intended application, ensuring compliance with engineering standards and safety requirements.

### Load Increment Summary

| Phase | Date | Time | Load (MT) | Pressure | Avg Settlement | Remarks |
|---|---|---|---|---|---|---|
| LOADING | 09/12/2025 | 11:29 AM | 0.00 | 0 kg/cm² | - mm | *Start* |
| LOADING | 09/12/2025 | 11:30 AM | 102.04 | 40 kg/cm² | 0.06 mm | - |
| LOADING | 09/12/2025 | 11:45 AM | 102.04 | 40 kg/cm² | 0.13 mm | - |
| LOADING | 09/12/2025 | 12:00 PM | 102.04 | 40 kg/cm² | 0.17 mm | - |
| LOADING | 09/12/2025 | 12:15 PM | 102.04 | 40 kg/cm² | 0.19 mm | - |
| LOADING | 09/12/2025 | 12:30 PM | 102.04 | 40 kg/cm² | 0.21 mm | - |
| LOADING | 09/12/2025 | 12:31 PM | 204.08 | 80 kg/cm² | 0.41 mm | - |
| LOADING | 09/12/2025 | 12:45 PM | 204.08 | 80 kg/cm² | 0.29 mm | - |
| LOADING | 09/12/2025 | 01:00 PM | 204.08 | 80 kg/cm² | 0.43 mm | - |
| LOADING | 09/12/2025 | 01:15 PM | 204.08 | 80 kg/cm² | 0.47 mm | - |
| LOADING | 09/12/2025 | 01:30 PM | 204.08 | 80 kg/cm² | 0.52 mm | - |
| LOADING | 09/12/2025 | 01:31 PM | 306.12 | 120 kg/cm² | 0.77 mm | - |
| LOADING | 09/12/2025 | 01:45 PM | 306.12 | 120 kg/cm² | 0.81 mm | - |
| LOADING | 09/12/2025 | 02:00 PM | 306.12 | 120 kg/cm² | 0.83 mm | - |
| LOADING | 09/12/2025 | 02:15 PM | 306.12 | 120 kg/cm² | 0.86 mm | - |
| LOADING | 09/12/2025 | 02:30 PM | 306.12 | 120 kg/cm² | 0.89 mm | - |
| LOADING | 09/12/2025 | 02:31 PM | 408.16 | 160 kg/cm² | 1.25 mm | - |
| LOADING | 09/12/2025 | 02:45 PM | 408.16 | 160 kg/cm² | 1.28 mm | - |
| LOADING | 09/12/2025 | 03:00 PM | 408.16 | 160 kg/cm² | 1.31 mm | - |
| LOADING | 09/12/2025 | 03:15 PM | 408.16 | 160 kg/cm² | 1.08 mm | - |
| LOADING | 09/12/2025 | 03:30 PM | 408.16 | 160 kg/cm² | 1.24 mm | *overide* |
| LOADING | 09/12/2025 | 03:31 PM | 510.20 | 200 kg/cm² | 1.51 mm | - |
| LOADING | 09/12/2025 | 03:45 PM | 510.20 | 200 kg/cm² | 1.50 mm | - |

| Phase | Date | Time | Load (MT) | Pressure | Avg Settlement | Remarks |
|---|---|---|---|---|---|---|
| LOADING | 09/12/2025 | 04:00 PM | 510.20 | 200 kg/cm² | 1.53 mm | - |
| LOADING | 09/12/2025 | 04:15 PM | 510.20 | 200 kg/cm² | 1.56 mm | - |
| LOADING | 09/12/2025 | 04:30 PM | 510.20 | 200 kg/cm² | 1.59 mm | - |
| LOADING | 09/12/2025 | 04:31 PM | 612.24 | 240 kg/cm² | 2.08 mm | - |
| LOADING | 09/12/2025 | 04:45 PM | 612.24 | 240 kg/cm² | 2.11 mm | - |
| LOADING | 09/12/2025 | 05:00 PM | 612.24 | 240 kg/cm² | 2.13 mm | - |
| LOADING | 09/12/2025 | 05:15 PM | 612.24 | 240 kg/cm² | 2.13 mm | - |
| LOADING | 09/12/2025 | 05:30 PM | 612.24 | 240 kg/cm² | 2.16 mm | - |
| LOADING | 09/12/2025 | 05:31 PM | 714.28 | 280 kg/cm² | 2.66 mm | - |
| LOADING | 09/12/2025 | 05:45 PM | 714.28 | 280 kg/cm² | 2.66 mm | - |
| LOADING | 09/12/2025 | 06:00 PM | 714.28 | 280 kg/cm² | 2.67 mm | - |
| LOADING | 09/12/2025 | 06:15 PM | 714.28 | 280 kg/cm² | 2.68 mm | - |
| LOADING | 09/12/2025 | 06:30 PM | 714.28 | 280 kg/cm² | 2.68 mm | - |
| LOADING | 09/12/2025 | 06:31 PM | 816.32 | 320 kg/cm² | 3.30 mm | - |
| LOADING | 09/12/2025 | 06:45 PM | 816.32 | 320 kg/cm² | 3.33 mm | - |
| LOADING | 09/12/2025 | 07:00 PM | 816.32 | 320 kg/cm² | 3.35 mm | - |
| LOADING | 09/12/2025 | 07:15 PM | 816.32 | 320 kg/cm² | 3.38 mm | - |
| LOADING | 09/12/2025 | 07:30 PM | 816.32 | 320 kg/cm² | 3.40 mm | - |
| LOADING | 09/12/2025 | 07:31 PM | 918.36 | 360 kg/cm² | 4.18 mm | - |
| LOADING | 09/12/2025 | 07:45 PM | 918.36 | 360 kg/cm² | 4.20 mm | - |
| LOADING | 09/12/2025 | 08:00 PM | 918.36 | 360 kg/cm² | 4.23 mm | - |
| LOADING | 09/12/2025 | 08:15 PM | 918.36 | 360 kg/cm² | 4.27 mm | - |
| LOADING | 09/12/2025 | 08:30 PM | 918.36 | 360 kg/cm² | 4.30 mm | - |
| LOADING | 09/12/2025 | 08:31 PM | 1020.40 | 400 kg/cm² | 4.95 mm | - |
| LOADING | 09/12/2025 | 08:45 PM | 1020.40 | 400 kg/cm² | 5.00 mm | - |
| LOADING | 09/12/2025 | 09:00 PM | 1020.40 | 400 kg/cm² | 5.03 mm | - |
| LOADING | 09/12/2025 | 09:15 PM | 1020.40 | 400 kg/cm² | 5.06 mm | - |
| LOADING | 09/12/2025 | 09:30 PM | 1020.40 | 400 kg/cm² | 5.08 mm | - |
| LOADING | 09/12/2025 | 09:31 PM | 1071.42 | 420 kg/cm² | 5.36 mm | - |
| HOLDING | 09/12/2025 | 10:30 PM | 1071.42 | 420 kg/cm² | 5.40 mm | - |
| HOLDING | 09/12/2025 | 11:30 PM | 1071.42 | 420 kg/cm² | 5.44 mm | - |
| HOLDING | 10/12/2025 | 12:30 AM | 1071.42 | 420 kg/cm² | 5.50 mm | - |

| Phase | Date | Time | Load (MT) | Pressure | Avg Settlement | Remarks |
|---|---|---|---|---|---|---|
| HOLDING | 10/12/2025 | 01:30 AM | 1071.42 | 420 kg/cm² | 5.53 mm | - |
| HOLDING | 10/12/2025 | 02:30 AM | 1071.42 | 420 kg/cm² | 5.58 mm | - |
| HOLDING | 10/12/2025 | 03:30 AM | 1071.42 | 420 kg/cm² | 5.60 mm | - |
| HOLDING | 10/12/2025 | 04:30 AM | 1071.42 | 420 kg/cm² | 5.62 mm | - |
| HOLDING | 10/12/2025 | 05:30 AM | 1071.42 | 420 kg/cm² | 5.66 mm | - |
| HOLDING | 10/12/2025 | 06:30 AM | 1071.42 | 420 kg/cm² | 5.67 mm | - |
| HOLDING | 10/12/2025 | 07:30 AM | 1071.42 | 420 kg/cm² | 5.68 mm | - |
| HOLDING | 10/12/2025 | 08:30 AM | 1071.42 | 420 kg/cm² | 5.74 mm | - |
| HOLDING | 10/12/2025 | 09:30 AM | 1071.42 | 420 kg/cm² | 5.79 mm | - |
| HOLDING | 10/12/2025 | 10:30 AM | 1071.42 | 420 kg/cm² | 6.14 mm | - |
| HOLDING | 10/12/2025 | 11:30 AM | 1071.42 | 420 kg/cm² | 6.63 mm | - |
| HOLDING | 10/12/2025 | 12:30 PM | 1071.42 | 420 kg/cm² | 6.86 mm | - |
| HOLDING | 10/12/2025 | 01:30 PM | 1071.42 | 420 kg/cm² | 6.95 mm | - |
| HOLDING | 10/12/2025 | 02:30 PM | 1071.42 | 420 kg/cm² | 7.00 mm | - |
| HOLDING | 10/12/2025 | 03:30 PM | 1071.42 | 420 kg/cm² | 7.12 mm | - |
| HOLDING | 10/12/2025 | 04:30 PM | 1071.42 | 420 kg/cm² | 7.17 mm | - |
| HOLDING | 10/12/2025 | 05:30 PM | 1071.42 | 420 kg/cm² | 7.19 mm | - |
| HOLDING | 10/12/2025 | 06:30 PM | 1071.42 | 420 kg/cm² | 7.21 mm | - |
| HOLDING | 10/12/2025 | 07:30 PM | 1071.42 | 420 kg/cm² | 7.22 mm | - |
| HOLDING | 10/12/2025 | 08:30 PM | 1071.42 | 420 kg/cm² | 7.22 mm | - |
| HOLDING | 10/12/2025 | 09:30 PM | 1071.42 | 420 kg/cm² | 7.22 mm | - |
| UNLOADING | 10/12/2025 | 09:31 PM | 1020.40 | 400 kg/cm² | 7.18 mm | - |
| UNLOADING | 10/12/2025 | 09:35 PM | 1020.40 | 400 kg/cm² | 7.18 mm | - |
| UNLOADING | 10/12/2025 | 09:45 PM | 1020.40 | 400 kg/cm² | 7.19 mm | - |
| UNLOADING | 10/12/2025 | 09:46 PM | 918.36 | 360 kg/cm² | 7.10 mm | - |
| UNLOADING | 10/12/2025 | 09:50 PM | 918.36 | 360 kg/cm² | 7.06 mm | - |
| UNLOADING | 10/12/2025 | 10:00 PM | 918.36 | 360 kg/cm² | 7.04 mm | - |
| UNLOADING | 10/12/2025 | 10:01 PM | 816.32 | 320 kg/cm² | 6.87 mm | - |
| UNLOADING | 10/12/2025 | 10:05 PM | 816.32 | 320 kg/cm² | 6.85 mm | - |
| UNLOADING | 10/12/2025 | 10:15 PM | 816.32 | 320 kg/cm² | 6.85 mm | - |
| UNLOADING | 10/12/2025 | 10:16 PM | 714.24 | 280 kg/cm² | 6.66 mm | - |
| UNLOADING | 10/12/2025 | 10:20 PM | 714.24 | 280 kg/cm² | 6.64 mm | - |

| Phase | Date | Time | Load (MT) | Pressure | Avg Settlement | Remarks |
|---|---|---|---|---|---|---|
| UNLOADING | 10/12/2025 | 10:30 PM | 714.24 | 280 kg/cm² | 6.63 mm | - |
| UNLOADING | 10/12/2025 | 10:31 PM | 612.24 | 240 kg/cm² | 6.42 mm | - |
| UNLOADING | 10/12/2025 | 10:35 PM | 612.24 | 240 kg/cm² | 6.37 mm | - |
| UNLOADING | 10/12/2025 | 10:45 PM | 612.24 | 240 kg/cm² | 6.35 mm | - |
| UNLOADING | 10/12/2025 | 10:46 PM | 510.20 | 200 kg/cm² | 6.14 mm | - |
| UNLOADING | 10/12/2025 | 10:50 PM | 510.20 | 200 kg/cm² | 6.13 mm | - |
| UNLOADING | 10/12/2025 | 11:00 PM | 510.20 | 200 kg/cm² | 6.13 mm | - |
| UNLOADING | 10/12/2025 | 11:01 PM | 402.16 | 160 kg/cm² | 5.89 mm | - |
| UNLOADING | 10/12/2025 | 11:05 PM | 402.16 | 160 kg/cm² | 5.87 mm | - |
| UNLOADING | 10/12/2025 | 11:15 PM | 402.16 | 160 kg/cm² | 5.84 mm | - |
| UNLOADING | 10/12/2025 | 11:16 PM | 306.12 | 120 kg/cm² | 5.58 mm | - |
| UNLOADING | 10/12/2025 | 11:20 PM | 306.12 | 120 kg/cm² | 5.54 mm | - |
| UNLOADING | 10/12/2025 | 11:30 PM | 306.12 | 120 kg/cm² | 5.51 mm | - |
| UNLOADING | 10/12/2025 | 11:31 PM | 204.08 | 80 kg/cm² | 5.25 mm | - |
| UNLOADING | 10/12/2025 | 11:35 PM | 204.08 | 80 kg/cm² | 5.21 mm | - |
| UNLOADING | 10/12/2025 | 11:45 PM | 204.08 | 80 kg/cm² | 5.18 mm | - |
| UNLOADING | 10/12/2025 | 11:46 PM | 102.04 | 40 kg/cm² | 4.89 mm | - |
| UNLOADING | 10/12/2025 | 11:50 PM | 102.04 | 40 kg/cm² | 4.81 mm | - |
| UNLOADING | 11/12/2025 | 12:00 AM | 102.04 | 40 kg/cm² | 4.80 mm | - |
| UNLOADING | 11/12/2025 | 12:01 AM | 0.00 | 0 kg/cm² | 4.14 mm | - |
| UNLOADING | 11/12/2025 | 12:05 AM | 0.00 | 0 kg/cm² | 4.10 mm | - |
| UNLOADING | 11/12/2025 | 12:15 AM | 0.00 | 0 kg/cm² | 4.06 mm | - |